Aadit Kshirsagar

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EDUCATION

University of Southern California Aug 2025 - May 2027
Master of Science in Computer Science *GPA:3.4/4*
Relevant Coursework*:* Foundations of AI, Machine Learning, Analysis of Algorithms, Natural Language Processing

Savitribai Phule Pune University Aug 2020 - Jun 2024
Bachelor of Engineering in Computer Science *GPA:3.61/4*
Relevant Coursework*:* Data Science, Cloud Computing, Deep Learning, Information Retrieval

TECHNICAL SKILLS

Programming: Python, Java, SQL, Git
Databases: PostgreSQL, MongoDB, Chroma, Pinecone, Milvus, Weaviate
App Development: HTML, CSS, JavaScript, Streamlit, FastAPI
Cloud and DevOps: Azure DevOps, Google Cloud, AWS, Docker, Kubernetes, Azure AI, OpenAI, Amazon Bedrock
AI/ML: Generative AI, Transformers, PyTorch, TensorFlow, NLP, Computer Vision, scikit-learn, Pandas, PySpark, MLOps, LLMOps, Prompt Engineering
LLM Frameworks: LangChain, LangGraph, LlamaIndex, Hugging Face, AutoGen, MCP

WORK EXPERIENCE

Tredence Inc | AI Intern | San Jose, CA Jun 2026 - Present
- Developing Generative AI solutions to optimize communication between Delivery Managers and customers across the Statement of Work and Purchase Order lifecycle.

Calsoft Pvt Ltd | Data Science Intern | Pune, India Apr 2024 - Jun 2025
- Led the end-to-end development of **enterprise GenAI systems**, from **system design** and implementation to **optimization** and **production deployment**.
- Built a multi-pipeline, **agent-based RAG system** that improved **test case retrieval relevance to 95%** while enabling **semantic search** across legacy test assets.
- Deployed an **on-premises AI solution** using open-source LLMs, **reducing manual test case generation by 80**% and accelerating QA workflows.

Intellect Design Arena Ltd | AI/ML Intern | Pune, India Sep 2023 - Mar 2024
- Engineered the **AI capabilities** of IGTB Copilot, enabling Relationship Managers to **extract actionable insights from banking data**.
- **Integrated LLMs and engineered prompt workflows** to deliver a Copilot proof of concept that **automated payment onboarding by over 70%**.

Intellect Design Arena Ltd | Software Developer Intern | Pune, India Jan 2023 - Mar 2023
- Engineered NLP pipelines to filter, structure, and standardize banking customer data for downstream KYC processing.
- Developed a Streamlit-based AI application that **automated** KYC verification workflows, **improving processing efficiency by 45%**.

PROJECTS

Medi Match Copilot | Codebase
- Built an **end-to-end multimodal clinical assistant** that combined **Llama-based NLP** and **medical image analysis** to automate symptom interpretation, imaging evaluation, and evidence-based diagnostic support.
- Engineered a **modular Python** architecture for **multimodal data retrieval**, hypothesis generation, and **vision-language reasoning**, enabling scalable, maintainable clinical decision support with up to **95% diagnostic accuracy**.

Eco Policy Copilot | Codebase
- Built a **LangGraph-based policy analysis platform** that **optimized** climate, economic, and energy policy recommendations by balancing emissions reduction targets against budget constraints.
- Engineered a **multi-agent workflow** for data ingestion, policy simulation, and executive summary generation, enabling environmental researchers to evaluate complex policy tradeoffs through evidence-based, cost-aware recommendations.

RESEARCH AND PUBLICATIONS

- Published research on **improving Retrieval-Augmented Generation using chunking and text splitting techniques** in the **International Journal of Scientific Research in Computer Science, Engineering, and Information Technology**. | Link

Research Assistant, USC Viterbi School of Engineering
- Built an **Alzheimer's disease classification** pipeline in **PyTorch** using OASIS brain MRI data, incorporating **ResNet-18 fine-tuning** and **multi-class evaluation** across four dementia stages.
- Engineered a robust training pipeline with **patient-level stratified splits**, **focal loss, class-balanced sampling**, and MRI-specific augmentations to mitigate class imbalance and data leakage, **achieving 90% classification accuracy**.